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11 July 1977
SUPERSEDES
MIL-P-48400 (MU)
8 May 1973

MILITARY SPECIFICATION

PROJECTILE, 60MM, HE, M720 METAL PARTS ASSEMBLY

This specification is approved for use by the U. S. Army Armament Research and Development Command, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers Projectile, 60MM, HE M720 Metal Parts Assembly.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

Military

MIL-I-6868	- Inspection Processes, Magnetic Particle
MIL-M-6867	- Magnetic Inspection Units
MIL-A-48078	- Ammunition, Standard Quality Assurance Provisions, General Specification for
MIL-C-48368	- Cartridge, HE, 60MM, XM720 Loading, Assembling and Packing

STANDARDS

Military

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-410	- Qualification of Inspection Personnel (Magnetic Particle and Penetrant)
MIL-STD-1167	- Ammunition Data Card

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, US Army Armament Research and Development Command, Attn. DRDAR-QA, Dover, New Jersey 07801 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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DRAWINGS

US Army ARRADCOM

D11751150	- Projectile, 60mm, HE, M720 Metal Parts Assembly
F-7548319	- Box Fiberboard Ammunition for Interplant Shipment of Projectile, Empty
9275526	- Cartridge, 60mm, HE, XM720

PUBLICATIONS

P.A. Manual No. 5-1	- Fragmentation Testing Procedures
MTP4-2-813	- Arena Test of High Explosive Fragmentation Munitions

(Copies of specifications, standards, drawings, and publications required by supplier in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications.- The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

ANSI-B46.1 - Surface Texture

(Application for copies should be addressed to the American National Standards Institute (ANSI), 1430 Broadway, New York, NY 10018.)

ASTM Method E-8	- Tension Test of Metallic Materials
ASTM Method E-10	- Brinell Hardness of Metallic Materials
ASTM Method E-18	- Rockwell Hardness and Rockwell Superficial Hardness of Metallic Materials
ASTM Method B-117	Salt Spray (Fog) Testing
ASTM Method A-322	- Hot-Rolled Alloy Steel Bars
ASTM Method E-350	- Chemical Analysis of Carbon Steel, Low-Alloy Steel, Silicon Electrical Steel, Ingot Iron and Wrought Iron

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(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

3. REQUIREMENTS

3.1 General Materials, parts, and assemblies shall comply with all requirements specified on drawing 11751150 and associated drawings and with all requirements specified in applicable specifications and standards. The materials specified on drawing 11751150 were selected to provide high fragmentation.

3.2 Stress Relief. Each projectile body shall be cold worked and stress relieved to the required physical properties.

3.2.1 Stress Relief. Each cold worked body shall be stress relieved at a minimum temperature of not less than 700 degrees (Farenheit) F. and a maximum temperature of not more than 725°F and held at temperature for a minimum of 30 minutes. No work involving plastic flow of metal is permitted after final stress relief.

3.3 Mechanical properties. All components shall meet the mechanical properties requirements as specified on the applicable drawings.

3.3.1 Hardness pattern. Projectile bodies shall be sectioned and a hardness pattern shall be taken to determine compliance with requirements of the applicable drawing.

3.3.2 Micro-Structure Test. The projectile bodies shall be sectioned and the micro-structure specimen shall meet the minimum of 90% spheroidization which is comparable to the degree of spheroidization depicted in Figure 3.

3.4 Metal defects (Magnetic particle inspection)

3.4.1 Body. The body, including the cavity shall be free from cracks, splits, bursts, cold shuts, piping, inclusions, folds, seams and other metal defects.

3.4.2 Body cavity. The body cavity shall be free of scale, fins, burns, draw marks, laminations, imbedded foreign matter, pits, porosity, and sharp edges.

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3.4.3 Visual standards. Visual standards for evaluation of metal defects (3.4.2) will be established for each contract after the first month's production or whenever requested by the procuring activity. (see 6.5) However, the establishment of visual standards shall not waive the requirement for compliance with 3.4.1 and 3.4.2.

3.5 Protective coating. Protective coating shall be in accordance with the applicable drawing and the specifications referenced thereon.

3.6 Excess Pressure/Service Pressure tests. The projectile metal parts assembly shall not breakup in the mortar tube or in flight at excess or service pressure.

3.7 Fragmentation tests

3.7.1 Arena test. (Applicable to first article samples only - For information). -The contractor shall submit projectile assemblies to the Government for Arena testing.

3.7.2 Pit test. The projectile assemblies shall comply with the requirements specified in Table I.

TABLE I

WEIGHT INTERVAL	FRAGMENT SIZE (GRAINS)	FRAGMENT QUANTITY	
		MINIMUM	MAXIMUM
A	2 - 3	327	479
B	4 - 6	254	338
C	10 - 15	96	165

3.8 Workmanship. The requirements for workmanship are as shown on the applicable drawings in the applicable specifications and the following:

3.8.1 Surface finish. The requirements for surface finish shall be as detailed on the applicable drawing.

3.8.2 Threads. Threads shall be full and undamaged for the entire minimum length as required by the applicable drawings.

3.8.3 Burr. No part shall have a burr which might interfere with the assembling of the round or which might be injurious to personnel handling the item.

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3.8.4 Foreign matter. No part or assembly shall contain dirt, grease, chips, rust, corrosion, or other foreign matter. Particular attention be given to the cavity of the body assembly, to insure that no grit, scale, or other foreign matter remains therein after final cleaning.

3.9 Process Description. As part of the process control document, DIP-1604 and at the time of first article submission, the contractor shall document and submit to the Government for review a description of his manufacturing process in sufficient detail to serve as a base line. The description shall specify all procedures critical to end item performance. Upon approval of the process control document and acceptance of the first article, the contractor shall not make changes to the manufacturing processes, materials used, drawings, specifications, or source of supply at stations determined by the Government to be critical to the product without prior approval from the Government. When the process control document is approved, the Government shall supply to the contractor a listing of those stations/operations considered critical. Changes to other than critical areas of the process control document shall not require Government approval, only notification. However, this does not relieve the contractor from the responsibility that significant changes may require an additional first article submission as determined by the Government. Tooling drawings, set up drawings, process parts drawings for all machining and forming operations up to and including nosing shall be submitted. A thermocouple chart of all thermal treatment operations shall be submitted. The thermocouple shall be inserted directly into the side wall of the piece and during thermal treatment continuous temperature and time readings shall be recorded.

3.10 First Article Inspection. This specification contains technical provisions for first article inspection. Requirements for the submission of first article samples by the contractor shall be as specified in the contract.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for Inspection and Standard Quality Assurance Provisions. Unless otherwise specified herein or in the contract, the provisions of MIL-A-48078 shall apply and are hereby made a part of this detail specification.

4.2 Classification of Inspections. The following types of inspection shall be conducted on this item:

a. First Article Inspection

b. Quality Conformance Inspection

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4.3 First Article Inspection

4.3.1 Submission. The contractor shall submit a first article sample and process description (see 3.9) as designated by the Contracting Officer for evaluation in accordance with provisions of 4.3.2. The first article sample shall consist of five (5) painted and five (5) unpainted projectile bodies in addition to the following quantities.

TEST	TEST PARAGRAPH	TEST QUANTITY
a. Excess pressure-service pressure test	4.5.4	50 complete projectile assemblies
b. Pit test	4.5.5.1	5 complete projectile assemblies
c. Arena Test	4.5.5.2	7 complete projectile assemblies

4.3.2 Inspections to be Performed. The First Article parts submitted in accordance with paragraph 4.3.1 will be inspected at a Government laboratory for all the requirements of the applicable drawings and specifications.

4.3.3 Rejection. See MIL-A-48078(FA) and in addition the sample submitted for size and quantity of fragments (Pit test) shall be tested as specified in 4.5.5.1. The lot shall be rejected if $\overline{X} \pm 1.3990\ s$ is greater than the upper limit or less than the lower limit at weight intervals A, B and C (see 3.7.2). Sample standard deviation shall be computed using the formula in 6.6. For additional criteria, see paragraphs 4.4.3.4 and 4.4.3.6.

4.4 Quality Conformance Inspection

4.4.1 Inspection Lot Formation. Inspection lots shall comply with the lot formation provisions of MIL-A-48078. In addition, inspection lots of projectile assemblies shall contain:

a. Stress relieved lot. Stress relieved lots shall contain cold worked bodies from not more than one heat of steel that have been cold worked by the same process utilizing the same tooling and equipment and spheroidized by the same cycle utilizing the same furnace, at the same temperature for the same length of time and stress relieved in the same furnace at the same temperature for the same length of time. The cold shaping lot thus constituted shall be the basis for the selection of samples for the tension, microstructure and hardness tests prescribed herein.

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b. Miscellaneous lots. Small lots, such as bodies from abnormally small heats of steel or bodies which have become separated from previously delivered lots, etc., may at the discretion of the manufacturer, be grouped into miscellaneous stress relieved lots. Such lots shall be identified by a miscellaneous lot code number and processed in accordance with (a) above. The miscellaneous lot thus constituted shall be the basis for the selection of samples for the mechanical properties testing hereinafter prescribed.

c. Projectile lots. When it has been determined that bodies of the stress relieved lot have complied with the mechanical properties requirement of the applicable drawing and specification, they may be grouped for further processing.

d. Shipping lots. Shipping lots shall consist of projectile assemblies produced by one supplier under one contract in one unchanged process, in accordance with the same drawings, revision, under conditions of a continuous production free of interruptions other than those due to the end of shift day or work week.

4.4.2 Examination. See MIL-A-48078. Unless otherwise specified in the Classification of Defects and Test tables, sampling plans for major and minor defects shall be in accordance with MIL-STD-105, Inspection Level II.

4.4.2.1 Tool control dimensions. Dimensions marked tool control (see drawing) shall be gaged at the beginning or end of each working day (24 hrs), during First Article Inspection and for the first component after each tooling change, adjustment or repair. Where destruction of the component is necessary to gage such dimensions, measurements of the tool may be substituted provided the contractor has established correlation between the tool dimensions and the component dimensions prior to the start of production. When acceptability has been determined, visual inspection of tool control dimensions will be permitted. Visual in operation. If any tool control dimension(s) is in doubt during visual inspection, the dimension(s) shall be immediately gaged to determine acceptability. Corrective action will immediately be instituted by the contractor whenever a defective tool control dimension is found.

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 1 OF 2		DRAWING NUMBER 11751150
4.4.2.2	Metal parts assembly, before painting				NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
CRITICAL					
1.	Metal defective		100%	3.4	Visual (Probe Light)
MAJOR					
101.	Weight		0.40%	3.1	Approved Scale
102.	Runout of pitch diameter of nose thread with bourrelet and front face		0.40%	3.1	Gage
103.	Runout of pitch diameter of rear thread with bourrelet and rear shoulder		0.40%	3.1	Gage
104.	Runout of ogive with bourrelet		0.40%	3.1	Gage
105.	Runout at rear of body with bourrelet		0.40%	3.1	Gage
106.	Runout of obturating ring groove with bourrelet		0.40%	3.1	Gage
107.	Diameter of bourrelet		0.40%	3.1	Gage
108.	Diameter of obturating ring groove		0.40%	3.1	Gage
109	Diameter of forward basic length on taper rear of bourrelet		0.40%	3.1	Gage
110.	Diameter at middle basic length on rear of bourrelet		0.40%	3.1	Gage

NOTES:

DRDAR-QA Form 160 Jul 77 Replaces SARPA-QA Form 2567 Feb 74 Which is Obsolete

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 2 OF 2		DRAWING NUMBER
4.4.2.2	Metal parts assembly, before painting				11751150
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
MAJOR					
111.	Diameter at rear basic length on taper near base end		0.40%	3.1	Gage
112.	Location of basic diameter on ogive (front)		0.40%	3.1	Gage
113.	Location of basic diameter on ogive (rear)		0.40%	3.1	Gage
114.	Thickness of wall ogive area		0.40%	3.1	Gage
115.	Thickness of wall at bourrelet		0.40%	3.1	Gage
116.	Thickness of wall at rear taper		0.40%	3.1	Gage
117.	Thickness through base		0.40%	3.1	Gage
118.	Width of obturating ring groove		0.40%	3.1	Gage
119.	Surface finish of cavity improper		0.40%	3.8.1	Visual (see 6.4)
120.	Foreign matter		0.40%	3.8.4	Visual
MINOR					
201.	Diameter of rear thread chamfer		0.65%	3.1	Gage
202.	Diameter of rear thread undercut (if applicable)		0.65%	3.1	Gage
203.	Length from front face to rear shoulder		0.65%	3.1	Gage
204.	Length of rear thread boss		0.65%	3.1	Gage
205.	Location of obturating ring groove		0.65%	3.1	Gage
206.	Width of rear thread undercut (if applicable)		0.65%	3.1	Gage
207.	Angle at rear of obturating ring groove		0.65%	3.1	Gage
208.	Radii or chamfers missing or incorrect		0.65%	3.1	Visual (see 6.4)
209.	Surface finish improper (except as otherwise classified)		0.65%	3.8.1	Visual (see 6.4)
210.	Burr		0.65%	3.8.3	Visual
211.	Foreign matter		0.65%	3.8.4	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 1 OF 2		DRAWING NUMBER
4.4.2.3	Metal parts assembly, after painting				11751150
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
CRITICAL					
1.	Metal defective		100%	3.4	Visual (Probe Light)
MAJOR					
101.	Pitch diameter of nose thread (max.)		0.40%	3.1	Gage
102.	Minor diameter of nose thread (max.)		0.40%	3.1	Gage
103.	Pitch diameter of nose thread (min.)		0.40%	3.1	Gage
104.	Pitch diameter of rear thread (min.)		0.40%	3.1	Gage
105.	Major diameter of rear thread (min.)		0.40%	3.1	Gage
106.	Pitch diameter of rear thread (max.)		0.40%	3.1	Gage
107.	Diameter of bourrelet, max.		0.40%	3.1	Gage
108.	Length of nose thread		0.40%	3.1	Gage
109.	Length of boss thread		0.40%	3.1	Gage
110.	Paint in cavity inadequate		0.40%	3.1	Visual
111.	Pool of paint in cavity		0.40%	3.1	Visual
112.	Thread(s) damaged or not full		0.40%	3.1	Visual
113.	Foreign matter		0.40%	3.8.4	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 2 OF 2		DRAWING NUMBER
4.4.2.3	Metal Parts assembly, after painting				11751150
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
MINOR					
201.	Corrosion preventive compound missing from threads and indicated surfaces		0.65%	3.1	Visual
202.	Paint on thread(s)		0.65%	3.1	Visual
203.	Paint not dry		0.65%	3.1	Visual/Tactile
204.	Paint inadequate or damaged		0.65%	3.1	Visual
205.	Marking missing, incorrect or illegible		0.65%	3.1	Visual

NOTES:

SARPA-QA FORM 2567 FEB 74 4810

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.4	Box, Fiberboard, prior to sealing				F7548319
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
CRITICAL	None defined				
MAJOR	None defined				
MINOR					
201.	Support missing or damaged		0.65%	3.1	Visual
202.	Separator missing or damaged		0.65%	3.1	Visual
203.	Improper assembly		0.65%	3.1	Visual
204.	Incorrect number in box		0.65%	3.1	Visual

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.5	Box, fiberboard, after sealing				F7548319
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
CRITICAL	None defined				
MAJOR	None defined				
MINOR					
201.	Box damaged, cut or gouged through all layers		0.65%	3.1	Visual
202.	Box not completely sealed		0.65%	3.1	Visual
203.	Contents loose		0.65%	3.1	Visual/Manual
204.	Mark missing, incorrect or illegible		0.65%	3.1	Visual

NOTES:

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4.4.3 Inspection testing. The following tests shall be performed in accordance with the provisions of 4.5.

4.4.3.1 Mechanical properties

4.4.3.1.1 Hardness test

4.4.3.1.1.1 Stress relieved lots. Twenty-five cold shaped bodies from the first batch of each cold shaping lot (as defined by 4.4.1.(a) processed through the stress relief furnace and five bodies from each hour's production thereafter for the balance of the lot shall be subjected to a hardness test specified in 4.5.1.1 with a Brinell Hardness Tester or by any other method approved by the procuring activity, to assess uniformity of the product. Hardness values shall be recorded.

4.4.3.1.1.2 Miscellaneous lots. Each miscellaneous lot shall be subjected to the hardness test in paragraph 4.4.3.1.1.1.

4.4.3.1.2 Tension test

4.4.3.1.2.1 Stress relieved lots. Two bodies, one of the hardest and one of the softest, as determined by the hardness test from the first twenty-five bodies selected in accordance with 4.4.3.1.1.1 shall be subjected to this test. Failure of any tension test specimen to comply with the elongation requirement of the applicable drawing shall be cause for rejection of the lot when tested as specified in 4.5.1.3. When it has been determined that the mechanical properties (elongation) comply and furnace temperature and time have been established, the furnace temperature and time shall be maintained throughout the stress relief lot. The furnace temperature shall not be less than the minimum temperature of 700°F nor greater than the maximum temperature of 725°F with a minimum soak time of 30 minutes. Furnace temperature and time will be recorded every hour to assess adequate control of stress relief equipment. The furnace temperature will be calibrated every eight hours by an independent method.

4.4.3.1.2.2 Miscellaneous lots. Samples for the tension test of miscellaneous stress-relieved lots shall be selected in accordance with 4.4.3.1.2.1. Failure of any tension test specimen to comply with the mechanical properties requirement of the applicable drawing shall be cause for rejection of the lot. Double the number of samples shall be selected for miscellaneous lots.

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4.4.3.1.3 Hardness pattern

4.4.3.1.3.1 Stress relieved lots. The samples supplied in accordance with 4.4.3.1.2.1 shall also have a Rockwell hardness pattern obtained. Failure to comply with the hardness requirements of the applicable drawing shall be cause for rejection of the lot when tested as specified in 4.5.1.2. When it has been determined that the hardnesses comply and furnace temperature and time have been established, the furnace temperature and time shall be maintained throughout the stress-relieved lot. The furnace temperature shall not be less than the minimum temperature of 700°F or greater than the maximum temperature of 725°F with a minimum soak time of 30 minutes. Furnace temperature and time will be recorded every hour to assess adequate control of stress relief equipment. The furnace temperature will be calibrated every eight hours by an independent method.

4.4.3.1.4 Microstructure Test

4.4.3.1.4.1 Stress-relieved lots. The samples selected in accordance to 4.4.3.1.2.1 shall be tested to determine the degree of spheroidization. Failure of the specimen to exhibit the degree of spheroidization depicted in the standard (see Figure 3) shall be cause for rejection of the stress relief lot when tested as specified in 4.5.1.4. When it has been determined that the microstructure specimen comply, the furnace temperature and time shall be established and maintained throughout spheroidization operation. An advisory spheroidization cycle is 1290°F at heat for 24 hours. At no time during the process shall the critical temperature of 1300°F maximum be exceeded in any thermal treatment after spheroidization. Furnace temperature and time shall be recorded every hour to assess adequate control of spheroidizing equipment. The furnace temperature shall be calibrated every eight hours by an independent method.

4.4.3.2 Metal defects (Magnetic particle inspection)

4.4.3.2.1 After stress relief and finish machining, bodies shall be inspected by magnetic particle inspection techniques approved by the responsible technical agency according to the following sequence when tested as specified in 4.5.2.

a. 100 percent magnetic particle inspection will be in affect until 2500 consecutive bodies have been inspected and found to be free of metal defects.

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b. Reduced inspection may then be instituted for magnetic particle inspection consisting of 30 bodies per hour for ten consecutive working days.

c. If no defect is found during the ten days, the sample may be reduced to 15 bodies per hour for five consecutive working days.

d. If no defect is encountered during the five days, the sample may be reduced to 5 bodies per hour.

e. If a defect is encountered by magnetic particle inspection during reduced sampling (b through d) the magnetic particle inspection shall revert back to (a) above. In addition a 100 percent back screening to the last acceptable hours production will be performed.

f. For requalification of reduced sampling, the manufacturer must follow the same procedures as outlined in paragraphs (a through d) above.

g. Should a controversy arise as to the validity of an indication, the manufacturer shall use metallographic test methods or an approved nondestructive method for proof of acceptability to the responsible technical agency.

h. Continual application of a 100 percent magnetic particle inspection by paragraph (a) above, shall not be used as an alternative procedure in lieu of corrective manufacturing process action to eliminate a continuing process induced defect. If corrective action is not instituted and the cause eliminated, the Government reserves the right to withdraw acceptance inspection until the contractor can prove his process is capable of producing suspect free projectiles.

4.4.3.3 Protective coating. The sample size and acceptance criteria shall be in accordance with the assembly drawing and the requirements of the specifications listed thereon. Failure to comply with the requirements of the applicable specifications shall be cause for taking corrective action recommended therein when tested in accordance with the test procedure in 4.5.3.

4.4.3.4 Excess pressure-service pressure tests. The following ballistic tests will be performed by the Government.

4.4.3.4.1 First article sample. The quantities listed in Table II shall be ballistically tested in the phases indicated.

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TABLE II

Phase	Quantity
Excess pressure	25
Service pressure	25

4.4.3.4.2 Production lot sample. The quantities listed in Table III shall be ballistically tested in the phases indicated.

TABLE III

Phase	Quantity
Excess pressure	10 each lot
Service pressure	10 each lot

4.4.3.4.3 Excess pressure. Samples provided in accordance with Tables II and III will be subjected to this test. Failure of any projectile assembly to comply with 3.6 shall be cause for rejection of the initial production sample or the production lot represented by the sample when tested as specified 4.5.4.1.

4.4.3.4.4 Service pressure. Samples provided in accordance with Tables II and III will be subjected to this test. Failure of any projectile assembly to comply with 3.6 shall be cause for rejection of the initial production sample or the production lot represented by the sample when tested as specified 4.5.4.2.

4.4.3.5 Fragmentation test (Pit test)

4.4.3.5.1 Beginning with the first lot produced and continuing until three consecutive lots have complied with the acceptance criteria specified, five (5) random sample projectiles (finished machined) shall be selected from each stress relieved lot for this test. The contractor shall ship the projectile bodies for loading and testing in accordance with instructions provided by the Contracting Officer (see 6.1). Prior to assembly the metal parts shall be weighed and permanently identified. The weights with round identifications shall be recorded and submitted as part of the test results. The projectile assemblies shall be loaded and completely assembled with fuze in accordance with drawing 9275526 and MIL-C-48368, The test shall be conducted and compliance to the requirement determined at weight intervals A and C (see 3.6.2). If $\bar{X} \pm 1.3990\ s$ is greater than the upper limits or less than the lower limits at the respective intervals the lot shall be rejected when tested as specified in 4.5.5.1. Sample standard deviation shall be computed using the formula in 6.6.

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4.4.3.5.2 After three consecutive lots have complied with the criteria of 4.4.3.5.1 three (3) random sample projectiles shall be selected from each stress relieved lot for this test. The contractor shall ship the projectile bodies for loading and testing in accordance with instructions provided by the Contracting Officer (see 6.1). Prior to assembly, the metal parts shall be weighed and identified. The weights with round identification shall be recorded and submitted as part of the test results. The projectile assemblies shall be loaded and completely assembled with fuze in accordance with drawing 9275526 and MIL-C-48368. The test shall be conducted and compliance to the requirement determined at weight intervals A and C (see 3.6.2). If $\bar{X} \pm 2.0000\ s$ is greater than the upper limit or less than the lower limit at the respective weight intervals, the lot shall be rejected when tested as specified in 4.5.5.1. Sample standard deviation shall be computed using the formula in 6.6.

4.4.3.6 Fragmentation test (Arena test). Seven (7) projectile assemblies shall be randomly selected and shipped for loading and testing in accordance with the instructions provided by the Contracting Officer (see 6.1). Prior to assembly, the metal parts shall be weighed and identified. The weights with round identification shall be recorded and submitted as part of the test results. The projectile assemblies shall be loaded and completely assembled in accordance with drawing 9275526 and MIL-C-48368. Results of test are for informational purposes. This test is performed on first article samples only.

4.4.3.7 Chemical composition. The sampling and acceptance criteria for determination of chemical composition shall be as specified in ASTM-A322 as required on the applicable drawing when tested as specified in 4.5.6

4.4.4 Inspection Equipment.The inspection equipment required to perform the examinations and tests prescribed herein is described in the 'Paragraph Reference/Inspection Method' column in the tables starting with paragraph 4.4.2.2. The contractor shall submit for approval inspection equipment designs in accordance with the terms of the contract. See Section 6 of MIL-A-43078.

4.5 Test methods and procedures.

4.5.1 Mechanical properties.

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4.5.1.1 Hardness test. Prior to finish machining, all bodies that will be subjected to the Brinell Hardness Test shall be prepared by grinding a flat on the cylindrical portion of the body where the obturator band groove will subsequently be machined. The test shall be performed using the equipment and procedures prescribed by ASTM Method E-10, except that the minimum dwell time shall be 5 seconds when a Brinell Hardness Test is used.

4.5.1.2 Hardness pattern.From the bodies selected for the tension test, a hardness pattern shall be obtained by removing a longitudinal section, at least one-half inch thick extending the length of the projectile body. The section shall be removed from approximately the center of the projectile where the centerline of the specimen is parallel with the walls of the specimen. The longitudinal walls shall be ground flat and parallel. A Rockwell hardness test shall be performed on one side of the ground surface 1/4 of an inch apart along the longitudinal axis (see Figure 1). Tension test specimen shall be taken from the bourrelet and boattail area. The hardness test shall be performed using the equipment and procedures prescribed by ASTM Method E-18.

4.5.1.3 Tension test.From each sample selected in accordance with 4.4.3.1.2, two tension tests (one bourrelet area, one boattail area) specimen conforming to the largest obtainable standard "R" type specimen as prescribed by ASTM Method E-8 shall be taken longitudinally from opposite sides of the body, 180 degrees apart. The number of specimens and their location is to determine the acceptability of the mechanical properties requirements, and with the approval of the procuring agency, the Government shall not be restricted from testing additional longitudinal specimens at other locations within the specified range, to determine that the mechanical properties (elongation) meet the drawing requirements. Procedures and criteria for determining yield strength, elongation, reduction in area and tensile strength shall be as prescribed by ASTM Method E-8, except that elongation shall be measured to the nearest thousandth (0.000) prior to conversion into percent elongation. Percent elongation shall be recorded to the nearest tenth (0.0).

4.5.1.4 Microstructure Test

4.5.1.4.1 Stress-relieved lots. A sample shall be selected for determining percent spheriodization. The sample shall be removed from an area between the upper and lower tension test specimen on one side of the projectile. The surface of the sample representing the longitudinal cross section shall be ground and polished for metallurgical examination. After polishing, the specimen shall be etched in nital reagent (97% ethyl alcohol - 3% nitric acid) to reveal the microstructure. The specimen shall be examined at 750-800

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magnification and compared with the standard (Figure 3) to determine degree of spheroidization. Figures 2 and 4 are provided for comparison purposes.

4.5.2 Metal defects (Magnetic particle inspection). The procedures for performing this test utilizing the wet flourescent method shall be as specified in MIL-I-6868, for indirect magnetization by either the continous method or the residual method, utilizing a circular magnetic field. The suspension shall be applied to the entire body, inside and outside. After draining, the body shall be thoroughly inspected in a darkened area under an ultra-violet light which will produce adequate flourescence of the magnetic particles. The solid content of the suspension shall be maintained at a level recommended by the suspension manufacturer. The suspension shall be checked at the start of each day's production and at intervals of not over 4 hours of continuous operation, utilizing the procedures and equipment prescribed by specification MIL-I-6868. When performing the test for solid content of the suspension, if it is determined that it does not comply, all projectile bodies that have been tested and accepted since the last satisfactory test was performed shall be re-tested after the suspension has been brought to the proper level. After acceptance, the body will be demagnetized and cleaned prior to stamping. The magnetic particle inspection shall be performed on equipment that complies with the requirements of MIL-M-6867. The operator(s) of the equipment and inspector(s) making the determination shall be certified in accordance with MIL-STD-410.

4.5.3 Protective coating.The procedures for testing the cleaning of components and the application of protective coating shall be in accordance with the requirements of the applicable drawing and the specifications listed thereon. The salt spray test shall be performed utilizing equipment prescribed by ASTM Method B117.

4.5.4 Excess pressure-service pressure tests

4.5.4.1 Excess pressure. The metal parts assemblies provided by Table II and III for this phase shall be inert loaded to the service weight and assembled with dummy fuzes into complete rounds. Firing tests will be conducted in a mortar for which the round is standard. The propellant charge will be adjusted to obtain 109 to 117 percent of the upper pressure limit of the propellant used with the applicable projectile. Photographic equipment will be set up near the gun muzzle to record the flight

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of the projectile to determine compliance with 3.6.1 on the initial production sample and on the first five production lot samples. An attempt will be made to recover at least 60 percent of the projectiles fixed in this phase. The recovered projectiles will be examined to determine compliance with 3.6.1.

4.5.4.2 Service pressure. The samples supplied in accordance with Tables II and III, inert loaded and fuzed will be assembled into complete rounds utilizing the service charge and fired in a mortar for which the round is standard for comparison with an equal number of rounds from a reference lot. The sample round and reference rounds will be fired alternately. The projectile will be observed and examined for compliance with 3.6.1. Muzzle velocities, range and deflection will be recorded for each round fired for determination of probable error (PE) in mils to assess range and deflection accuracy. Data obtained will be recorded for information only and forwarded to the congnizant technical agency.

4.5.5 Fragmentation tests

4.5.5.1 Pit test. Prior to testing, the fuze shall be modified for static initiation by replacing the safe and arming mechanism with an adapter and inserting a No. 6 or No. 8 Electric Blasting Cap. The test shall be performed in accordance with the procedure specified in P.A. Manual No. 5-1 (see 6.7). The fragments shall be counted, weighed and grouped to determine compliance with the requirement.

4.5.5.2 Arena test. The arena test shall be conducted in accordance with the procedures specified in Material Test Procedure (MTP) 4-2-813.

4.5.6 Chemical composition. The chemical composition shall be determined in accordance with the test procedures specified in ASTM-A322 referencing ASTM-E350.

5. PREPARATION FOR DELIVERY

5.1 Packaging, packing and marking. Packaging, packing and marking shall be in accordance with requirements of Drawing F7548319.

5.2 Data cards. Data card information shall be as specified in MIL-STD-1167.

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6. NOTES

6.1 Ordering Date. See MIL-A-48078(PA) and in addition procurement document should specify Government Facility and Government Proving Ground for tests in paragraph 4.4.3.4.

6.2 Submission of Designs for Approval. See 6.2.3 of MIL-A-48078. DRDAR-QAR-I, Dover, New Jersey 07801.

6.3 Data requirements. Deliverable data required by this specification is cited in the following paragraphs:

Paragraph	Data Requirement	Applicable DIP
3.9	Process Description	DIP 1604

(Copies of data item descriptions required by the contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer).

6.4 Visual examination qualifications. When compliance with the applicable classification of defect is in doubt as a result of visual examination, this characteristic may be measured or gaged to determine acceptability.

6.5 Visual standards. Visual standards to be established at contractor's plant in accordance with 3.4.3.

6.6 Calculation of sample standard deviation. The sample standard deviation will be calculated using the following formula or its equivalent:

$$s = \sqrt{\frac{\Sigma (x - \bar{x})^2}{n - 1}}$$

$\bar{x}$ = Average value for the sample
x = Numerical value for individual observations
s = Sample standard deviation
n = Sample size

6.7 Submission of first article sample. Instructions as to the location for the evaluation of the first article sample shall be obtained from the contracting officer. Upon receipt of each request, the contracting officer shall advise ARRADCOM and instructions will be issued accordingly. All inquiries should be forwarded to:

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Commander
ARRADCOM
ATTN: DRDAR-QAR-Q
Dover, New Jersey 07801

6.8 Proving Ground Test Summary

TEST	SAMPLE SIZE	REQUIREMENT
First Article Test		
(a) Excess pressure-service pressure test	50 projectile assemblies	3.6
(b) Pit Test	5 projectile assemblies	3.7.2
(c) Arena Tests	7 projectile assemblies	for information (3.7.1)
Regular Production		
(a) Excess pressure-service pressure test	20 projectile assemblies	3.6
(b) Pit Test	5 projectile assemblies (This sample is applicable to the first lot produced and continuing until three (3) consecutive lots have complied with the requirement.)	3.7.2
	3 projectile assemblies (This sample applies after three consecutive lots have complied with the requirement.)	3.7.2

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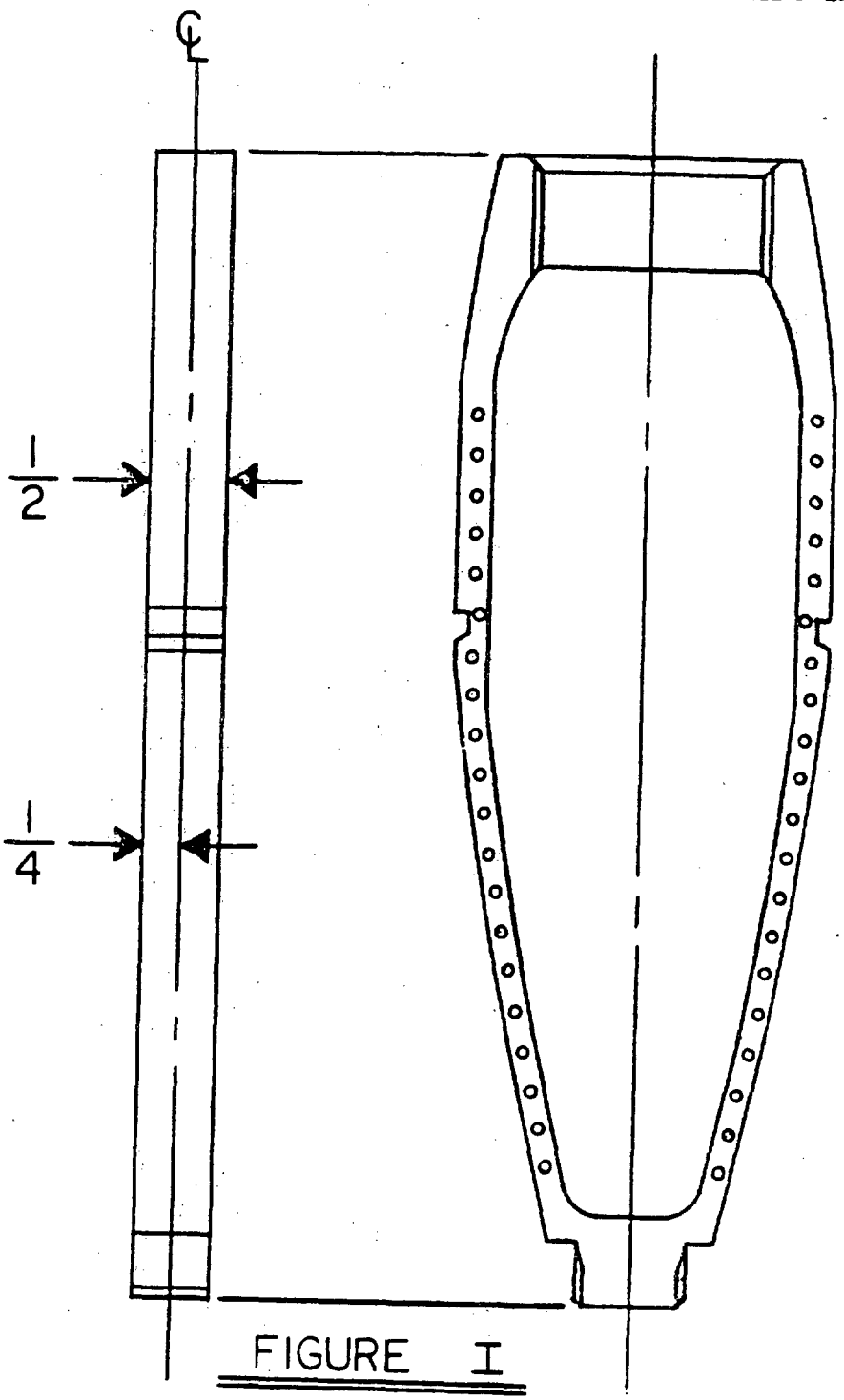


FIGURE I

HARDNESS PATTERN

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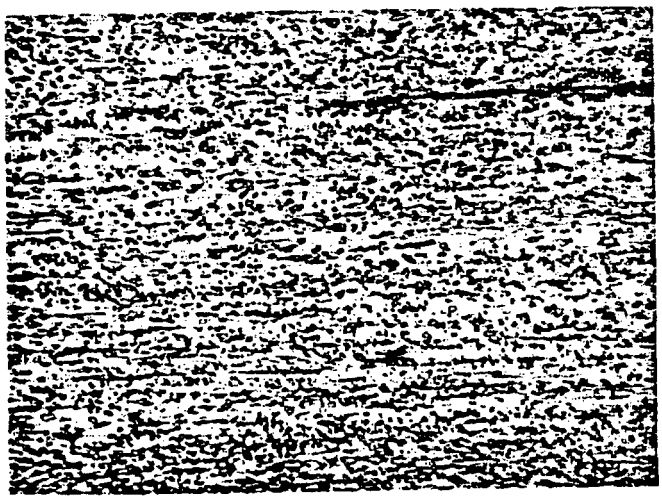


Figure 2 - Complete Spheroidization

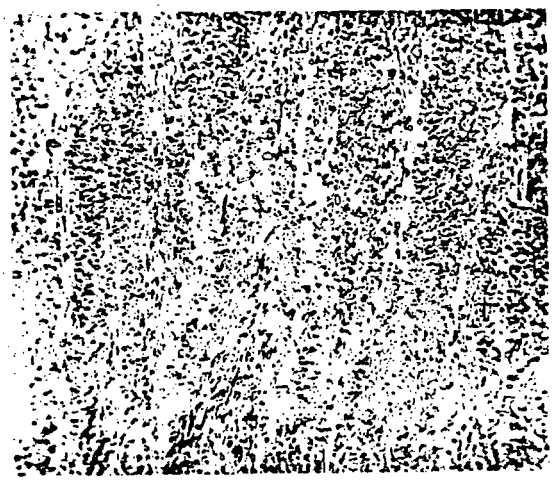


Figure 3 - 90% Spheroidization (minimum)



Figure 4 - Unacceptable Spheroidization

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